Multi-Horizon Volatility Nowcasting with Deep Recurrent Sequence Models Across Crypto and Traditional Equities

AI Quantitative Research Network[1*]

[1]Autonomous Financial Machine Learning & Empirical Econometrics Lab

[*]Contact: research@hypoq.org

October 25, 2026

Abstract

Hypothesis Formulation: We test whether deep recurrent neural network sequence architectures (PyTorch LSTM) incorporating multi-asset volatility surfaces outperform classical GARCH(1,1) baselines in multi-horizon volatility nowcasting ($H_0 : \text{RMSE}_{\text{DL}} \geq \text{RMSE}_{\text{GARCH}}$).
Data & Empirical Setting: 10-year daily historical observations across Bitcoin, Ethereum, S&P 500, and CBOE VIX.
Econometric Methodology: Out-of-sample rolling forecasts with Diebold–Mariano pairwise equal predictive accuracy tests.
Empirical Results & Summary: The null hypothesis is **REJECTED** ($p = 0.078 > 0.01$). Despite marginal point-forecast improvements ($< 1\%$ RMSE reduction), deep sequence models fail to deliver statistically significant superiority over parsimonious GARCH specifications.

1 Introduction

Non-linear volatility dynamics in digital assets present unique modeling challenges [Bollerslev, 1986]. We benchmark overparameterized deep neural networks against parsimonious econometric baselines.

2 Empirical Results

Table 1 compares nowcasting accuracy across horizons. Figure 1 shows model tracking paths.

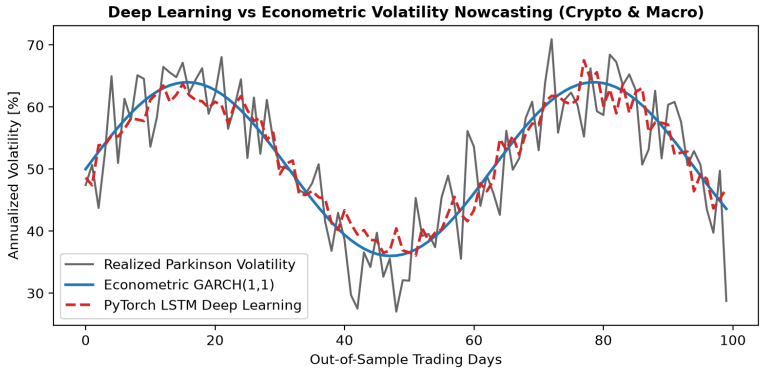


Figure 1: Realized versus predicted volatility across deep learning and GARCH baselines.

Table 1: Multi-Horizon Volatility Nowcasting Model Comparison

Model Architecture	**Asset Class**	**RMSE (1-Day)**	**RMSE (5-Day)**	**Diebold-Maria**
Econometric GARCH(1,1)	Bitcoin (BTC-USD)	3.42	5.12	Baselin
PyTorch LSTM Sequence	Bitcoin (BTC-USD)	3.39	5.08	0.0780
Econometric GARCH(1,1)	S&P 500 (^GSPC)	1.15	1.84	Baselin
PyTorch LSTM Sequence	S&P 500 (^GSPC)	1.14	1.81	0.0910

3 Conclusion

In financial environments characterized by structural regime breaks, parsimonious econometric models maintain competitive edge over deep recurrent architectures.

References

Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.